Network Coverage

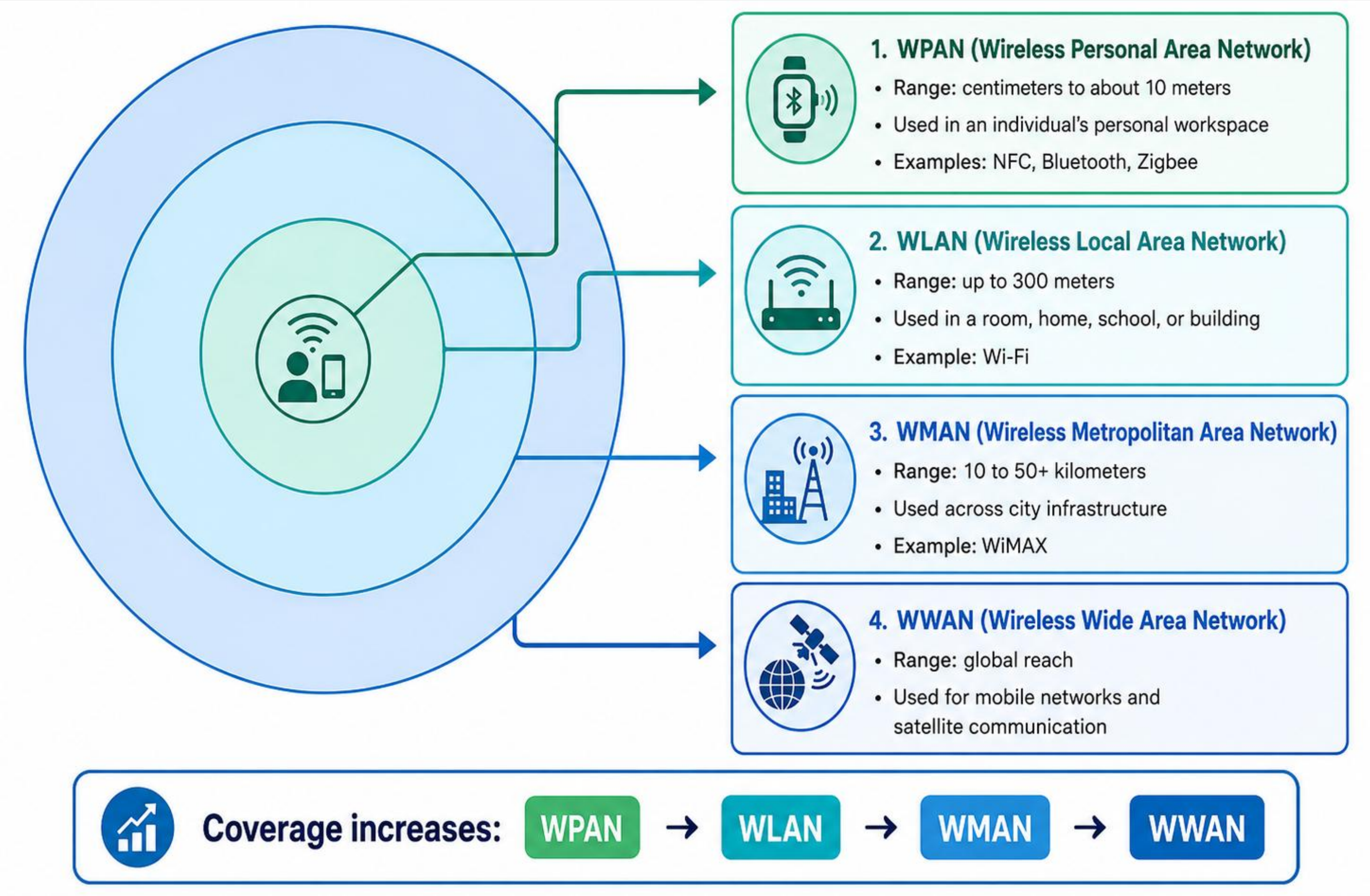
WPAN
Personal
3–10 m

WLAN
Local
50–300 m

WMAN
Metropolitan
10–50 km

WWAN
Wide
Country/World

Network Coverage



NFC: Near Field Communication

Full form: Near Field Communication.

Very short-range communication: usually a few centimeters.

Works when two devices are almost touching/tapping.

Used in debit/credit cards, security passes, toll/transport cards and quick information sharing.

NFC = Tap to connect

Short range gives better security

Example: card payment or metro pass

NFC: Near Field Communication

Zigbee: Low Power IoT Communication

IEEE Standard: 802.15.4.

Used in low-powered wireless devices and IoT systems.

Low data rate, low bandwidth and low power consumption.

Examples: smart lights, temperature sensors, smoke detectors, smart meters, industrial automation.

Zigbee: Low Power IoT Communication

Zigbee vs NFC

Feature	Zigbee	NFC
Main idea	Low-power IoT/sensor network	Very short-range tap communication
Range	Short/room-level sensor network	Few centimeters
Use	Smart home, sensors, automation	Payment, ID/pass, transport card
Best phrase	Low power, low bandwidth	Tap and go

Bluetooth: Short-range Personal Connection

Wireless technology for communication between nearby devices.
Works around 2.45 GHz frequency band.

Network type: WPAN / Personal Area Network.

Common uses: earbuds, speakers, mouse, keyboard, smartwatch, file sharing.

IEEE Standard: 802.15

Typical range:
3–10 meters

Low power
Lower speed

Half Duplex Communication

Bluetooth: Short-range Personal Connection

Technical Dashboard

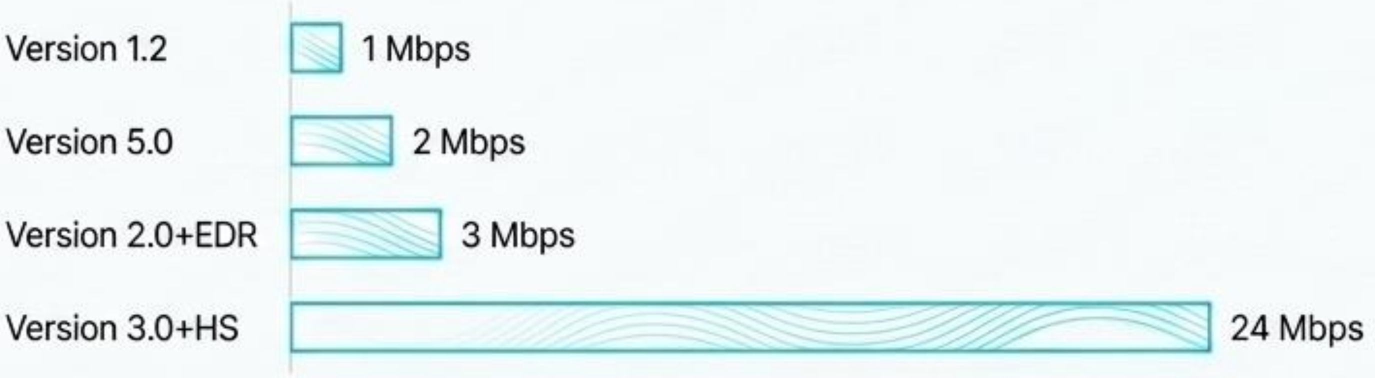
Standard
IEEE 802.15

Frequency
2.45 GHz

Base Range
3 to 10 meters (up to 100m with power modifications)

Transmission Mode
Half-duplex

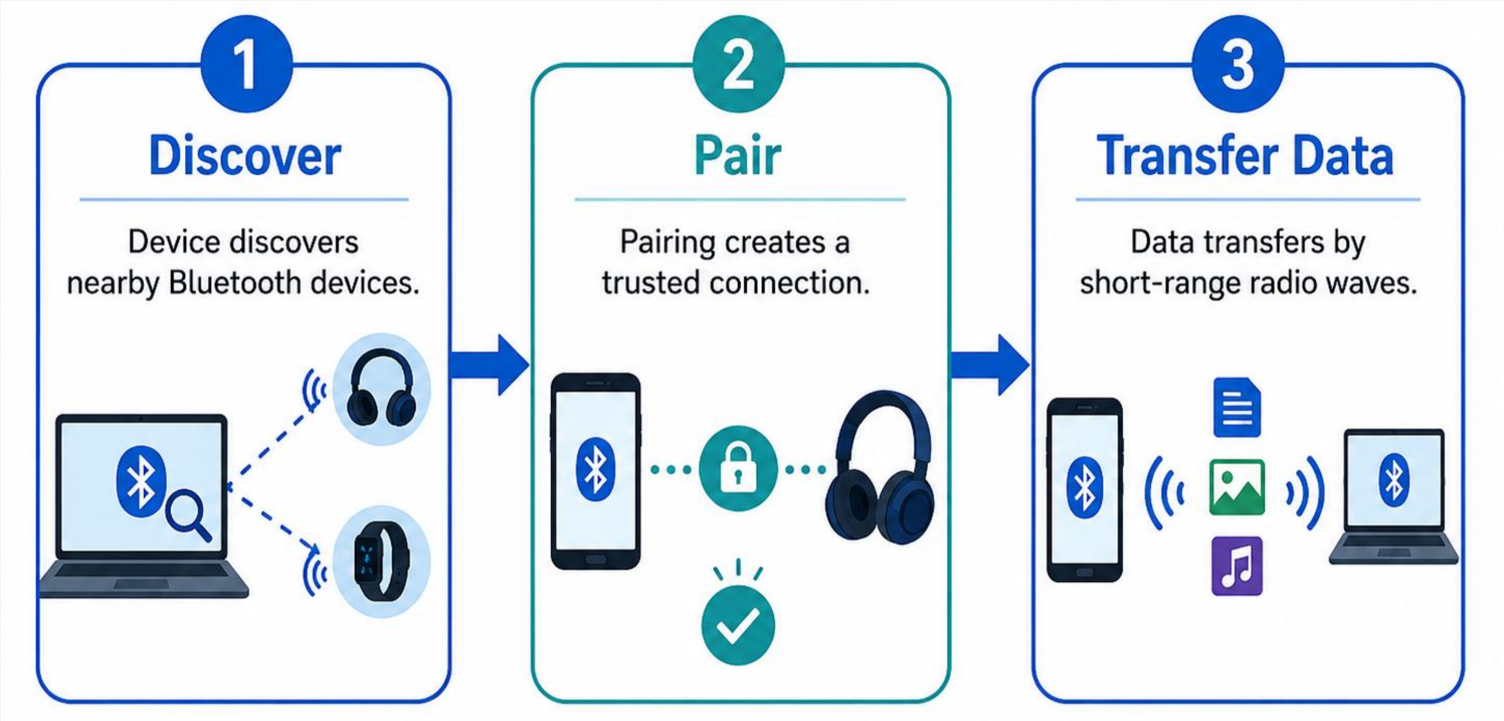
Evolution of Speed



Historical Callout Box

Invented in 1994 by Ericsson as an RS-232 cable alternative. Named to honor the 10th-century Danish King "Harald Bluetooth" Gormsson, who united various tribes—mirroring the technology's goal to unite different communication protocols.

How Bluetooth Works: Pairing

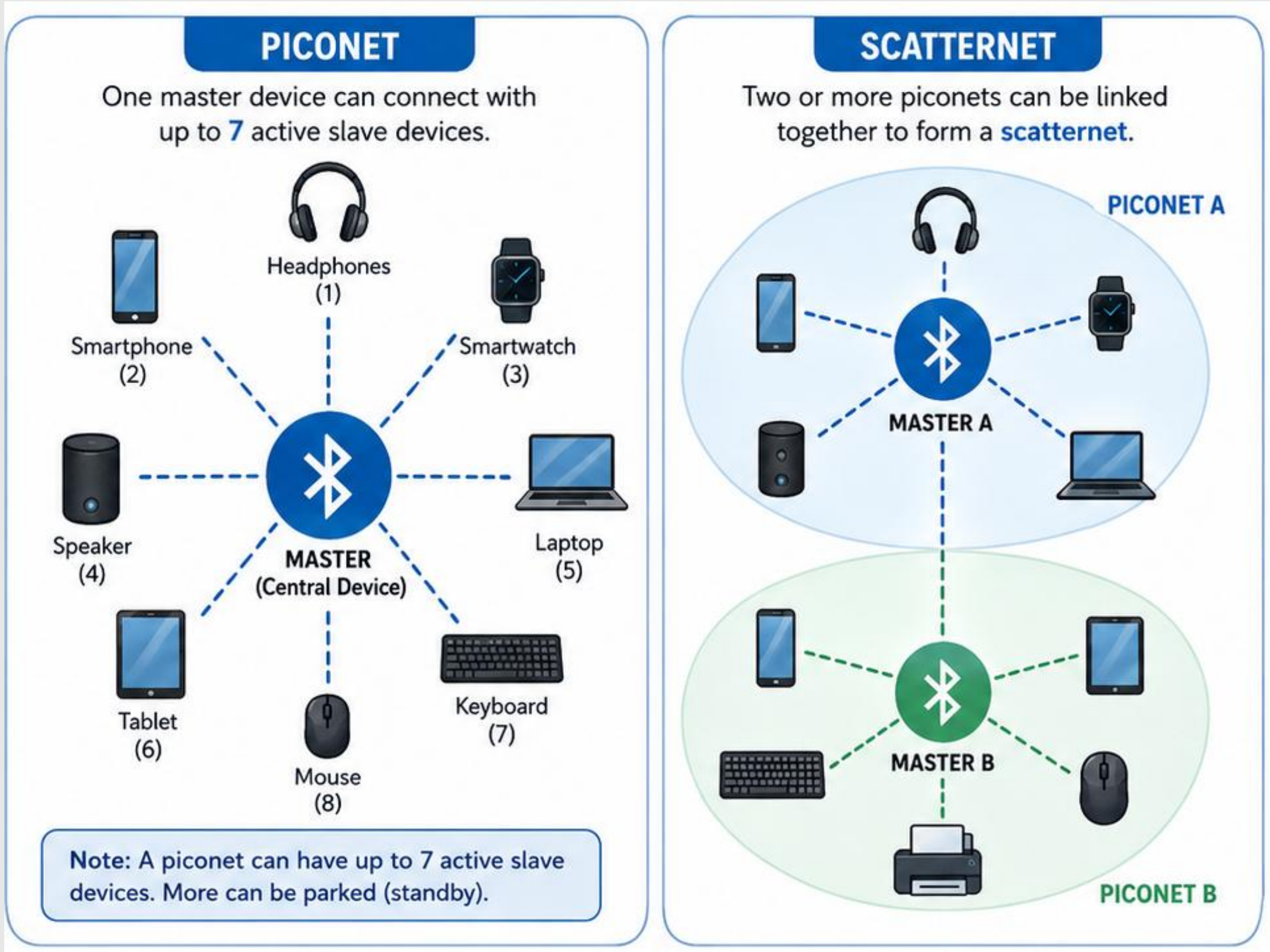


Step 1: Device discovers nearby Bluetooth devices.

Step 2: Pairing creates a trusted connection.

Step 3: Data transfers by short-range radio waves.

How Bluetooth Works: Piconet and Scatternet



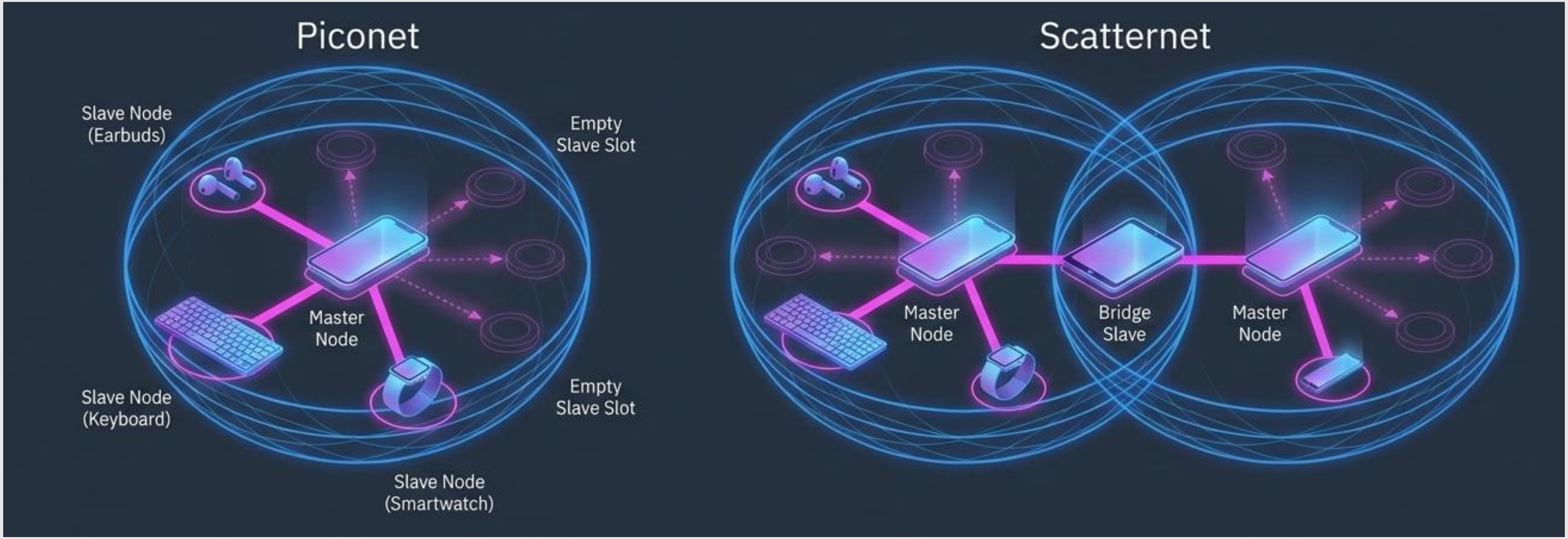
Piconet: Master device connects nearby slave devices
Multiple piconets = Scatternet

Bluetooth is like talking to a friend sitting beside you.

A Bluetooth small network is called a piconet.

Multiple piconets can create a scatternet.

How Bluetooth Works: Piconet and Scatternet



Wi-Fi: Wireless Local Area Network

Wi-Fi is used to create a Wireless Local Area Network (WLAN).

Connects phones, laptops, tablets and smart TVs to a router/access point.

Usually works in 2.4 GHz and 5 GHz bands.

Used at home, school, office, café, airport and public hotspot.

IEEE Standard: 802.11

Network type: WLAN

Range: about 50–300 m

WAP

Internet reaches the router through cable/fiber/mobile backhaul.

Router broadcasts wireless radio signals.

Your device sends requests back to the router.

The router brings data from internet and sends it to your device.

Wi-Fi Security and Hotspot

Wi-Fi is easy to connect, so security is important.

Use strong password and encryption.

A hotspot is a fixed wireless internet coverage area.

Hotspots are common in schools, cafés, airports and libraries.

Discussion: What makes a Wi-Fi password weak?

Example: 12345678 is easy to guess.

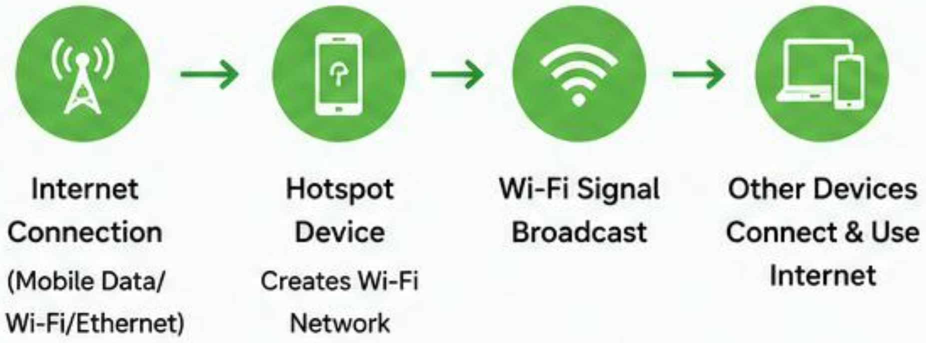
Wi-Fi Security and Hotspot

1 WHAT IS A HOTSPOT?

A hotspot is a feature that allows a device (smartphone, laptop, or dedicated device) to share its internet connection with other devices via Wi-Fi.

2 HOW DOES IT WORK?

The hotspot device uses mobile data or another internet connection and broadcasts a Wi-Fi signal. Other devices connect to this signal to access the internet.



3 TYPES OF HOTSPOT

Mobile Hotspot

Smartphone uses mobile data to share internet.

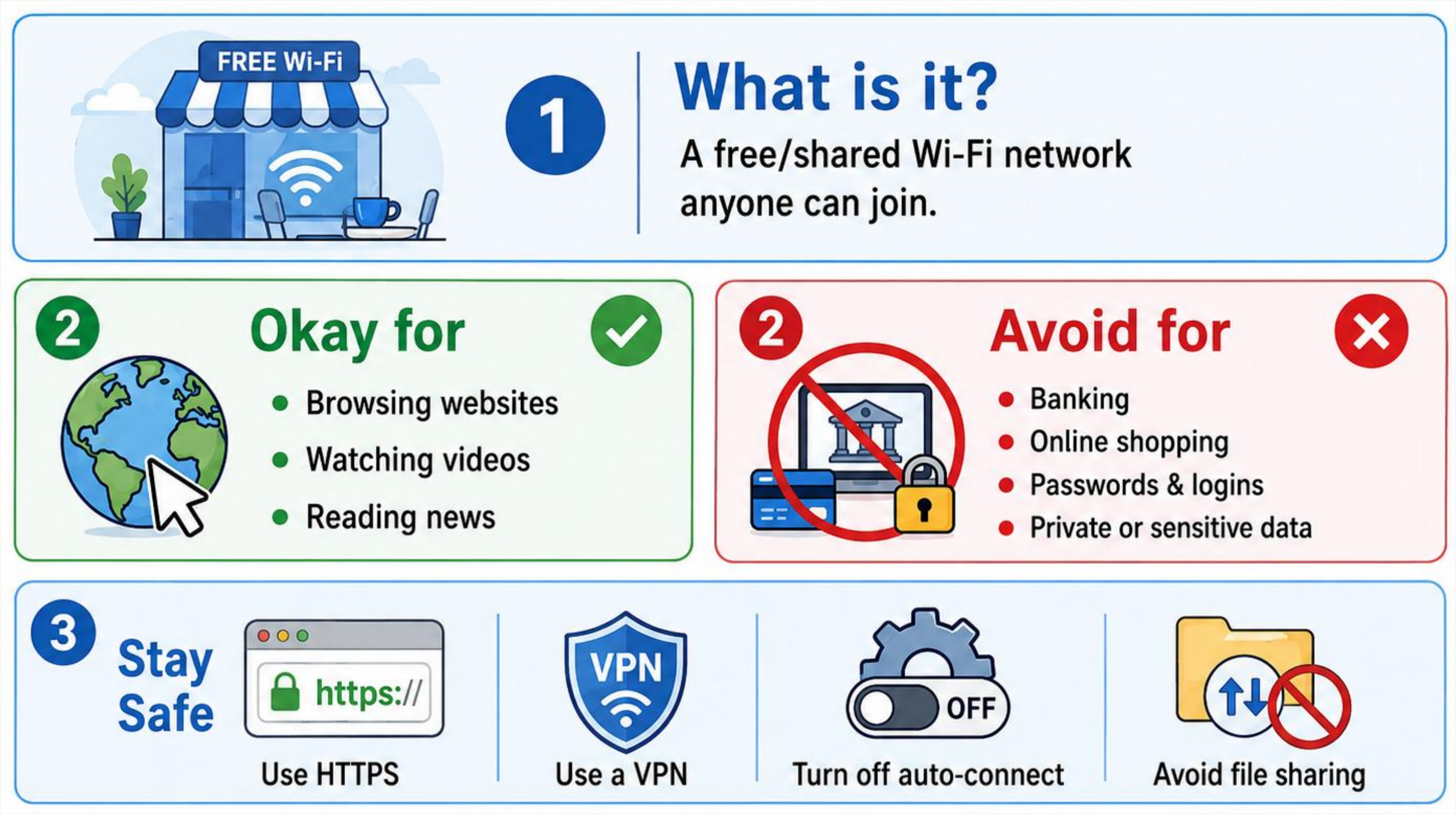
Wi-Fi Hotspot (Laptop)

Laptop connects to internet and shares via Wi-Fi.

Dedicated Hotspot Device

Portable devices made for sharing internet.

Public or Open Wi-Fi: Should we use?



WiMAX: Broadband Over a Large Area

Full form: Worldwide Interoperability for Microwave Access.

A high-speed wireless broadband technology.

Network type: WMAN / Wireless Metropolitan Area Network.

Useful for rural/remote areas where cable internet is difficult.

IEEE Standard: 802.16

Coverage: 10–50 km

Needs base station + receiver

WiMAX: Broadband Over a Large Area

Technical Dashboard

Standard	Base Coverage	Data Rate	Transmission Mode
IEEE 802.16 (WiMAX)	10 to 50 km (up to 70 km)	80 Mbps up to 1+ Gbps	Full-duplex

Core Components Architecture

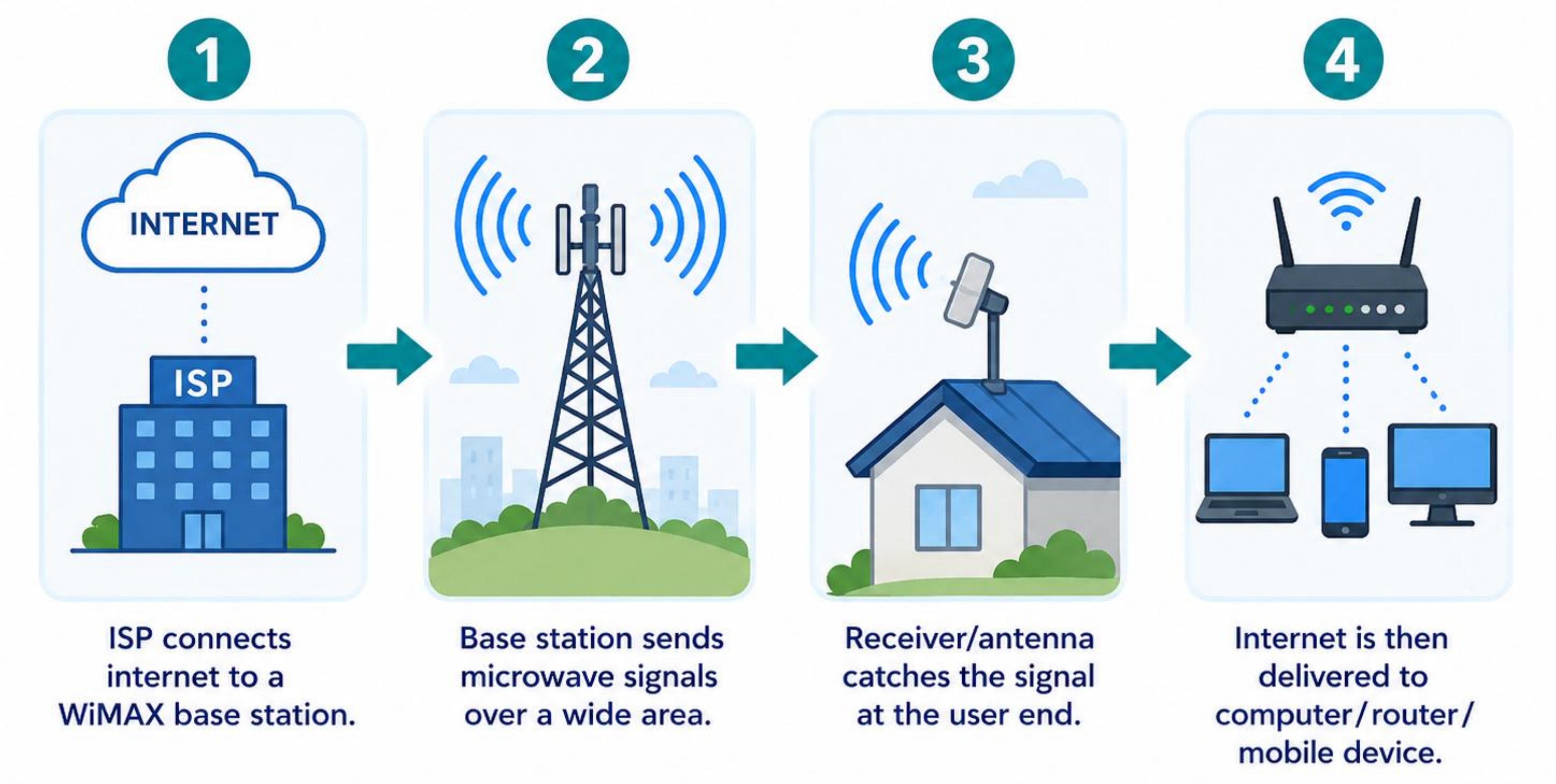
Base Station (Tower)

Broadcasts over massive geographical areas to thousands of clients. Operates similarly to cell towers.

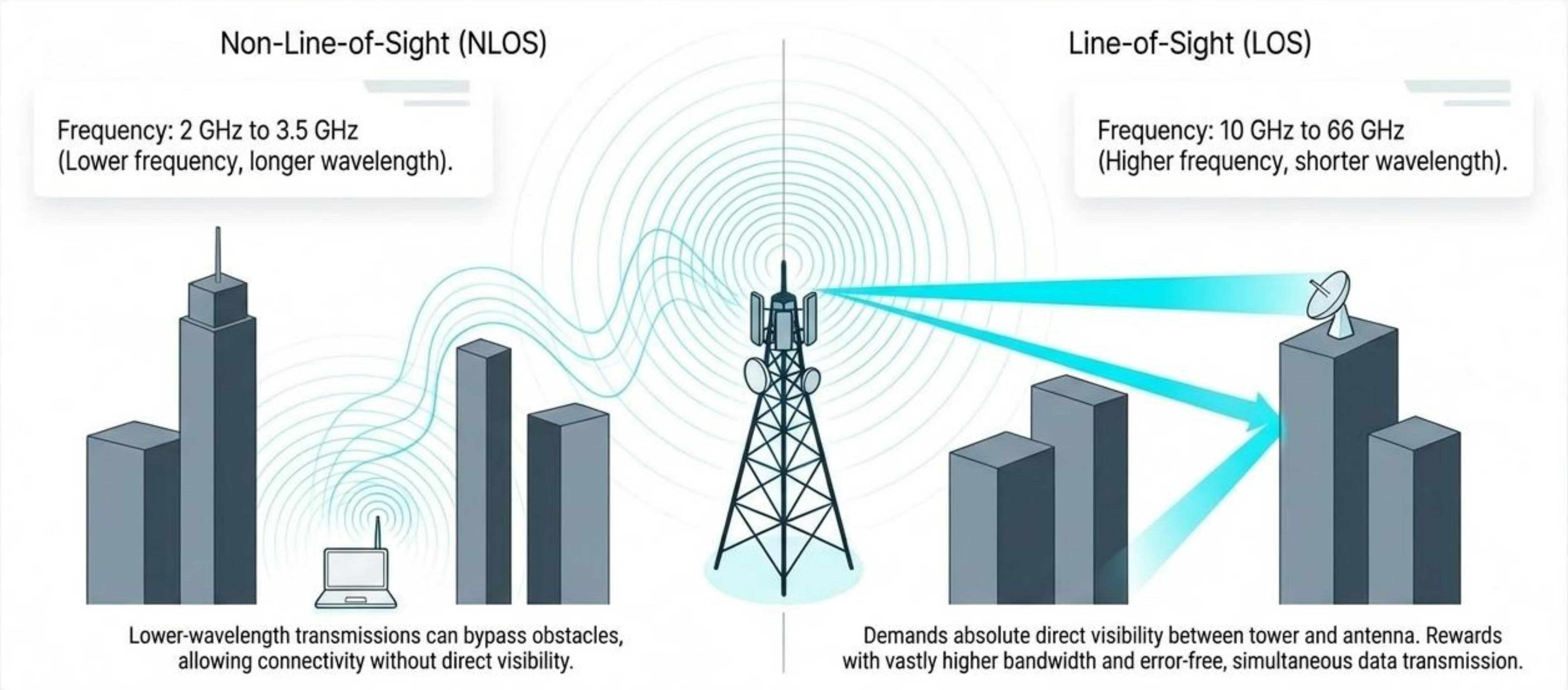
WiMAX Receiver

Built into devices or standalone modems provided by ISPs, featuring a dedicated antenna.

WiMAX: Broadband Over a Large Area

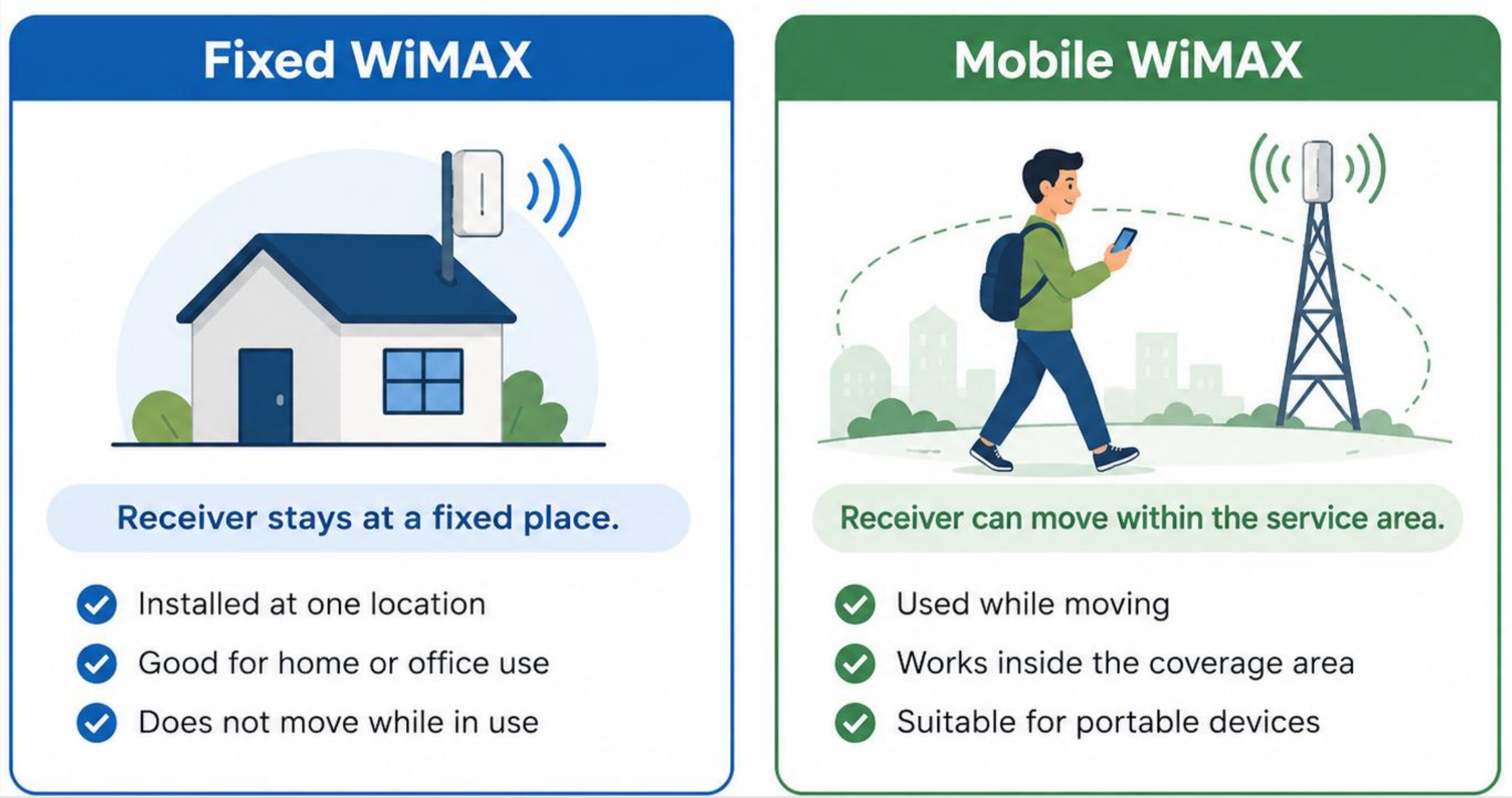


WiMAX: LOS, NLOS



WiMAX: LOS, NLOS

WiMAX: Fixed and Mobile



Comparison: Bluetooth vs Wi-Fi vs WiMAX

Feature	Bluetooth	Wi-Fi	WiMAX
IEEE Standard	802.15	802.11	802.16
Network Type	WPAN	WLAN	WMAN
Frequency	2.4–2.45 GHz	2.4–5 GHz	2–66 GHz
Range	3–10 m	50–300 m	10–50 km
Speed	Low	Medium–High	High
Main Use	Personal devices	Home/school internet	Wide-area broadband